Open Name Tags (ONT)

A short, human-readable name — like `alice` — that you truly own. design brief · v0 prototype

Most names online are really *accounts*: a company hands them out, and can rename you, reclaim them, or shut you down. An ONT name is different — it is controlled by a cryptographic key that only you hold. No company is involved, there is nothing to renew or pay rent on, there is no token, and no one (not even ONT's creators) can move or revoke it. Anyone can look up who owns a name and confirm the answer for themselves.

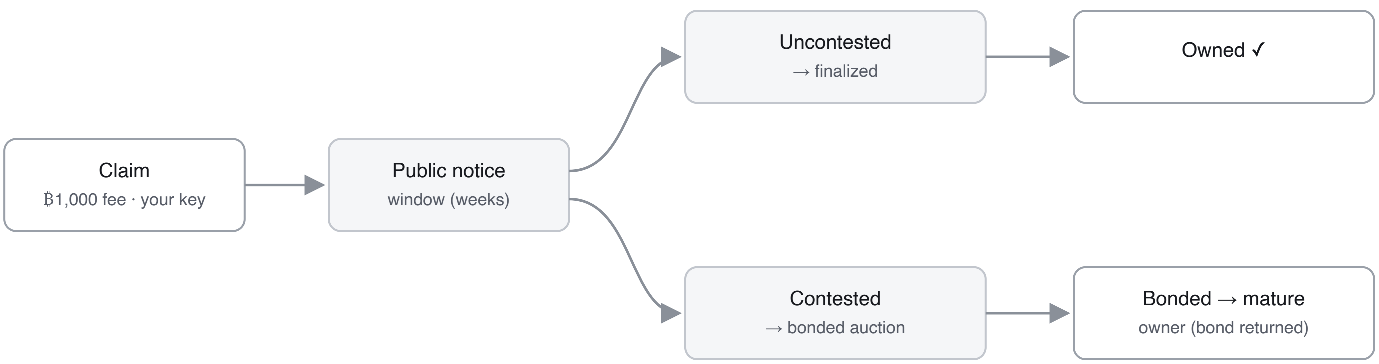
WHAT YOU'D USE ONE FOR

- **A payment handle** — pay `alice` instead of a long address; the name points to wherever she wants to be paid.
- **An identity handle** — one username for open-source / decentralized apps that no platform can reassign or revoke.
- **Addressing for services or software agents** (early).

HOW YOU GET A NAME

There is one path for every name. It forks only if two people want the same one.

1. **Claim it.** Pay a small, one-time Bitcoin fee — about $1 (**₿1,000**) — paid to Bitcoin's miners, not to ONT.
2. **A public notice window opens** (a few weeks). This is everyone else's chance to contest the name — to put in a competing claim if they want it too. (A competing claim doesn't take the name; it forces the auction in step 4.)
3. **If no one else does, the name is yours.** Thousands of these uncontested claims are bundled into one small Bitcoin record — which is what keeps it cheap across billions of names.
4. **If someone else also claims it, the name is *contested*** — and only then is it settled by an auction: each bidder locks up bitcoin as a **returnable bond** (they keep custody; it is released later), and the largest bond wins. Genuinely contested names are rare, so auctions are the exception.



A second claim during the notice window is what makes a name contested; the auction is then decided by the largest returnable bond, not by who claimed first. Most names are never contested — they finalize when the window closes.

Either way, you end up with the same thing: a globally unique name controlled by your key.

WHAT OWNING A NAME LETS YOU DO

A name is controlled by one key — your **owner key**. With it you can:

- **Point the name somewhere** — at a Bitcoin or Lightning address, a website, and so on — and change it whenever you like. (These mappings live off-chain, signed by your key, so updates are instant, free, and never touch Bitcoin.)
- **Transfer** the name to someone else's key.
- **Set up recovery** ahead of time, so a lost key isn't the end — and only the backup key you chose can use it, so recovery can never become a way for someone to take your name.

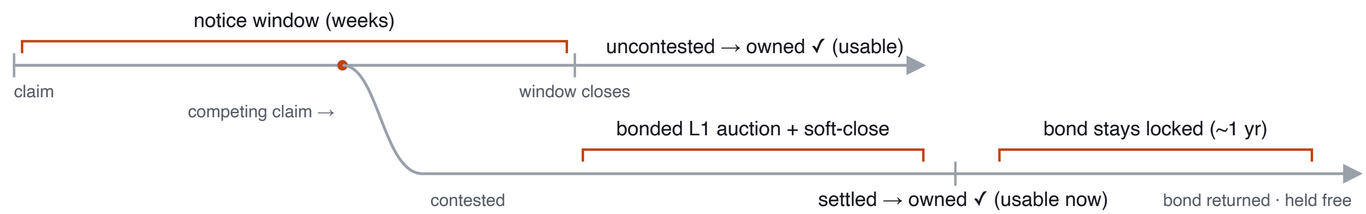
WHAT IT COSTS BITCOIN

Claiming isn't free chain-spam. The fee you pay covers the Bitcoin transaction your claim rides in — a batch only counts if its miner fee is at least the sum of the claims inside it — so each name buys the blockspace it uses. And that space is tiny: about **0.016 virtual bytes** per name once batched (one ~150-byte anchor carries ~10,000 claims).

WHY YOU CAN TRUST IT

No company, server, or founder decides who owns a name — Bitcoin does. The rules that turn Bitcoin transactions into ownership live in a small, **frozen program (about seven files)** that anyone can audit. Run it over Bitcoin's history and you get the same answer everyone else does, and you can check that answer against Bitcoin's own block headers and proof-of-work — so a server that lies about who owns a name gets caught, not believed. The services that help you find and publish names — *resolvers* — only mirror this data; they never decide it.

THE NUMBERS WE'RE PROPOSING — SEVERAL ARE PLACEHOLDERS, ALL OPEN TO CHALLENGE



Either path gives you a usable, owned name — uncontested at window close, contested the moment the auction **settles**. For a contested name you can use and transfer it right away; the bond just stays locked through maturity (~1 yr) and then returns, after which the name is held with no bond at all.

PARAMETER	PROPOSED	STATUS
Claim fee (every name)	**Ƀ1,000** ~$1, sunk, to miners	baseline
Contested-auction min bond	**Ƀ50,000** ~$50, returnable	placeholder
Bond maturity	~52,560 blocks (~1 yr)	test override
Notice window	weeks, height-keyed	placeholder · fairness lever
Data-availability windows	unset	deadline for batch bytes to surface + reorg depth; unpinned
On-chain footprint	~0.016 vB/name; fee = Σ fees	measured

Opening bond for scarce short names.

NAME LENGTH	OPENING BOND	~USD
1 char	**Ƀ100,000,000** (1 BTC)	~$100k
2 char	Ƀ50,000,000	~$50k
3 char	Ƀ25,000,000	~$25k
4 char	Ƀ12,500,000	~$12.5k
5+ char	fee only; Ƀ50,000 floor if contested	~$1 / ~$50

Only the very short (≤4-char) names carry a high length-scaled opening bond (halving per added character); everything else just uses the flat fee plus a bond if contested.

Least sure of: the **contest rate** is unknown until launch — we assume it's high early (everyone wants `bitcoin`, dictionary words) and low for the long tail (`sallysmith2165`); and the notice window has to be long enough that real owners can contest a day-one land-rush.

STATUS — HONEST (MATURITY, NOT DIRECTION)

Live on a Bitcoin test network (signet), end-to-end: claim, owner-key transfer, owner-signed records, recovery, and a bonded auction bid the resolver accepts — with the consensus code and signatures cross-checked byte-for-byte against a second independent implementation. **Prototype / not yet wired:** the cheap batched-claim path into the live indexer (built and unit-tested, including convergence against a data-withholding adversary); a single-writer publisher; and producers don't yet emit the proofs a phone/browser would check. Not mainnet-ready.

WHAT WE MOST WANT YOU TO PUSH ON

- **Data availability + convergence** — is the fail-closed, height-keyed rule (a batch counts only if its bytes surface by a deadline) sound against reorgs and withholding? Is the right signal an on-chain availability marker, or pure timing?
- **On-chain footprint** — are the ≤135-byte OP_RETURN events acceptable on mainnet, or is a script/covenant carrier worth a soft-fork dependency?
- **Light-client verification** — a launch blocker, or fine post-launch?
- **Auction form** — open ascending vs. sealed second-price, given MEV and relay-bid timing?
- **Launch fairness** — is a long notice window enough against a day-one rush on premium names, or do we need a decaying launch fee?

Repo: github.com/deekay/ont · the full design brief, prior-art comparison, risk register, and the verification you can run yourself: `docs/ONT_DESIGN_BRIEF.md`.